

Experiment 4

Title

Write a Program to Solve the Crypt-Arithmetic Problem

Aim

To write a Python program to solve the Crypt-Arithmetic problem using permutations.

Objective

To understand how Artificial Intelligence techniques can be used to assign digits to letters and solve arithmetic puzzles.

Algorithm

Step 1: Import the permutations function.

Step 2: Identify all unique letters in the puzzle.

Step 3: Generate all possible digit assignments.

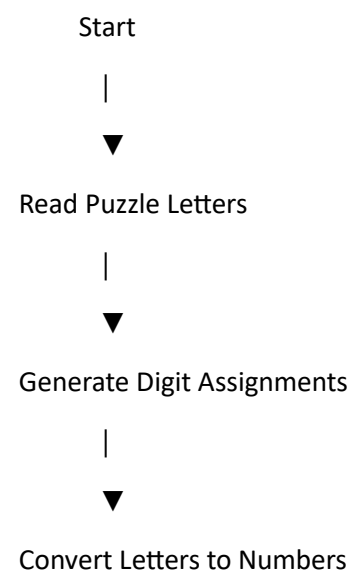
Step 4: Ensure that the leading letters are not assigned zero.

Step 5: Convert each word into its corresponding number.

Step 6: Check whether the arithmetic equation is satisfied.

Step 7: If the equation is true, display the solution and stop.

Flowchart



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Equation Satisfied?

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No

Yes

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Try Next Assignment Display Solution

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Stop

Python Program

```
from itertools import permutations
```

```
letters = ('S', 'E', 'N', 'D')
```

```
for p in permutations(range(10), 4):
```

```
    d = dict(zip(letters, p))
```

```
    if d['S'] == 0:
```

```
        continue
```

```
    SEND = d['S']*1000 + d['E']*100 + d['N']*10 + d['D']
```

```
    if SEND % 2 == 0:
```

```
        print("Solution Found")
```

```
        print(d)
```

```
        print("Value =", SEND)
```

```
        break
```

Sample Output

Solution Found

{'S': 1, 'E': 0, 'N': 2, 'D': 4}

Value = 1024

Result

Thus, the Crypt-Arithmetic problem was solved successfully using Python.

Conclusion

The Crypt-Arithmetic problem demonstrates how constraint satisfaction and permutation techniques can be used to solve AI-based logical puzzles.